

FACTORY REALLOCATION & SHIPPING OPTIMIZATION RECOMMENDATION SYSTEM FOR NASSAU CANDY DISTRIBUTOR

ABSTRACT

In today's competitive supply chain environment, efficient logistics and factory allocation play a significant role in reducing operational costs and improving customer satisfaction. Nassau Candy Distributor manufactures and distributes multiple confectionery products through different production facilities across the United States. The existing factory allocation system follows static business rules that often lead to inefficient shipping routes, higher delivery lead times, and increased logistics costs.

This project presents an intelligent Factory Reallocation and Shipping Optimization Recommendation System using Machine Learning and Data Analytics. The system analyzes historical sales and shipment data to identify inefficiencies in the existing factory assignment process. A predictive machine learning model estimates shipping lead time, while a recommendation engine evaluates alternative factory assignments and suggests the most efficient production location for each product.

The project includes comprehensive data preprocessing, feature engineering, exploratory data analysis, predictive modeling, K-Means clustering for route analysis, and an interactive Streamlit dashboard for visualization. The dashboard enables users to analyze factory performance, compare current and recommended allocations, perform what-if scenario analysis, and evaluate operational improvements.

The developed system demonstrates how machine learning can assist organizations in making data-driven supply chain decisions that improve shipping efficiency while maintaining profitability.

Keywords: Machine Learning, Streamlit, Supply Chain Optimization, Factory Allocation, Logistics Analytics, Linear Regression, K-Means Clustering, Recommendation System.

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CHAPTER 1 – INTRODUCTION

Supply chain management has become one of the most important aspects of modern businesses. Efficient transportation and factory allocation directly influence delivery time, customer satisfaction, and operational cost. Companies that manufacture products at multiple factories often face challenges in deciding which factory should produce which product for a particular customer region.

Traditionally, organizations use fixed allocation rules that assign products to predefined factories regardless of changing demand, transportation distance, or regional requirements. While this approach is simple to implement, it often results in inefficient shipping routes, higher transportation costs, and increased delivery lead time.

The advancement of Artificial Intelligence, Machine Learning, and Data Analytics provides organizations with new opportunities to optimize these decisions using historical data. Instead of relying on static rules, predictive models can estimate delivery performance under different scenarios and recommend better factory allocations.

This project develops a Machine Learning-based Factory Reallocation and Shipping Optimization Recommendation System for Nassau Candy Distributor. The system predicts shipping performance and recommends alternative factory assignments that improve operational efficiency without negatively affecting profitability.

The solution integrates predictive analytics, recommendation logic, clustering techniques, and an interactive Streamlit dashboard, allowing business users to evaluate multiple scenarios before implementing operational changes.

CHAPTER 2 – COMPANY BACKGROUND

Nassau Candy Distributor is a confectionery distribution company that manufactures and distributes a wide variety of chocolate, sugar, and novelty candy products through multiple manufacturing facilities.

The company operates several factories located in different geographical regions. Each factory specializes in manufacturing particular products. Customer orders are received from multiple regions, requiring products to be transported from factories to destination locations.

Currently, factory allocation is based on fixed business rules rather than dynamic optimization. As business volume increases, this strategy becomes less efficient because transportation distance, regional demand, and logistics performance continuously change over time.

This project aims to improve the company's operational efficiency by introducing a data-driven recommendation system capable of suggesting better factory allocations.

CHAPTER 3 – PROBLEM STATEMENT

3.1 Background

Nassau Candy Distributor manufactures and distributes multiple confectionery products through several production facilities. Each product is assigned to a predefined factory based on static business rules. While this approach is simple, it does not consider changing customer demand, transportation distance, factory workload, or regional shipping efficiency.

As the number of customer orders increases, static allocation becomes inefficient because the optimal factory for a product may vary depending on the destination region. This results in longer delivery times, unnecessary transportation costs, and reduced profitability.

Traditional allocation systems are unable to simulate different factory assignment scenarios before implementation. Consequently, management lacks sufficient decision-support tools for evaluating alternative production strategies.

3.2 Problem Definition

The current factory allocation process suffers from several operational limitations:

- Products are assigned to factories using fixed rules.
- Shipping distance is often not minimized.
- Delivery lead time increases for several customer regions.
- Transportation cost negatively affects overall profitability.
- No intelligent recommendation system exists to suggest better factory assignments.
- Management cannot evaluate the operational impact of changing factory allocations before implementation.

3.3 Research Problem

The primary research problem addressed by this project is:

How can machine learning and data analytics be used to recommend better factory assignments that reduce shipping lead time while maintaining profitability?

3.4 Need for the Project

The existing business process provides descriptive information regarding orders and shipments but does not provide actionable recommendations for improving operational efficiency.

A decision-support system is required that can:

- Predict shipping performance
- Compare multiple allocation scenarios
- Recommend the best production facility
- Improve logistics efficiency

- Reduce delivery lead time
 - Support managerial decision making
-

CHAPTER 4 – PROJECT OBJECTIVES

The major objectives of this project are:

Primary Objectives

- Develop an intelligent factory recommendation system.
 - Predict shipping lead time using Machine Learning.
 - Identify inefficient shipping routes.
 - Recommend optimal factory allocation for each product.
 - Improve overall logistics efficiency.
 - Reduce delivery lead time.
 - Maintain business profitability.
-

Secondary Objectives

- Perform exploratory data analysis.
 - Clean and preprocess raw business data.
 - Engineer useful machine learning features.
 - Analyze factory-wise performance.
 - Cluster shipping routes using K-Means clustering.
 - Perform What-If scenario analysis.
 - Develop an interactive Streamlit dashboard for decision support.
-

Business Objectives

The developed system aims to provide the following business benefits:

- Faster deliveries
 - Reduced transportation cost
 - Better utilization of manufacturing facilities
 - Improved customer satisfaction
 - Data-driven managerial decisions
 - Operational efficiency improvement
-

CHAPTER 5 – EXISTING SYSTEM

5.1 Current Working Process

Currently, Nassau Candy Distributor allocates products to manufacturing facilities using predefined business rules. Every product is permanently associated with a particular factory regardless of changing transportation conditions or customer demand.

The existing workflow consists of the following steps:

1. Customer places an order.
2. Product is assigned to its default factory.
3. Factory manufactures the product.
4. Product is shipped to the customer.
5. Delivery is completed.

Although simple, this process ignores several important operational factors.

5.2 Limitations of Existing System

The existing allocation strategy has several shortcomings:

Static Allocation

Each product has only one assigned factory, preventing flexible production planning.

Long Shipping Routes

Some customers receive products from distant factories even when another facility could deliver faster.

Increased Transportation Cost

Long-distance shipping increases logistics expenditure.

Poor Resource Utilization

Some factories remain underutilized while others become overloaded.

Lack of Predictive Analytics

The system cannot estimate future shipping performance.

No Recommendation Capability

The existing process provides no intelligent suggestions regarding better factory assignments.

Absence of Decision Support

Management cannot evaluate multiple allocation strategies before implementation.

CHAPTER 6 – PROPOSED SYSTEM

6.1 Overview

The proposed Factory Reallocation and Shipping Optimization Recommendation System introduces an intelligent decision-support framework using Machine Learning and Business Analytics.

Instead of assigning products using fixed business rules, the proposed system evaluates historical shipment data, predicts delivery performance, and recommends the most suitable factory for each product.

6.2 Working Methodology

The proposed system performs the following operations:

Step 1 – Data Collection

Historical sales, shipping, customer, product, and factory data are collected from the Nassau Candy Distributor dataset.

Step 2 – Data Preprocessing

Raw data undergoes:

- Missing value handling
 - Duplicate removal
 - Data type conversion
 - Date processing
 - Feature generation
-

Step 3 – Feature Engineering

New analytical features are generated, including:

- Shipping Lead Time
 - Factory Coordinates
 - Customer Coordinates
 - Shipping Distance
 - Encoded categorical variables
-

Step 4 – Machine Learning Prediction

A Linear Regression model predicts shipping lead time using operational variables.

Step 5 – Recommendation Engine

Alternative factory assignments are simulated.

The system compares:

- Current factory
 - Recommended factory
 - Predicted lead time
 - Operational improvement
-

Step 6 – K-Means Route Clustering

Shipping routes are grouped according to operational similarity to identify inefficient logistics patterns.

Step 7 – Streamlit Dashboard

A user-friendly dashboard provides:

- Executive Dashboard
 - Factory Performance Analysis
 - Recommendation Engine
 - What-If Scenario Analysis
 - Route Clustering Dashboard
-

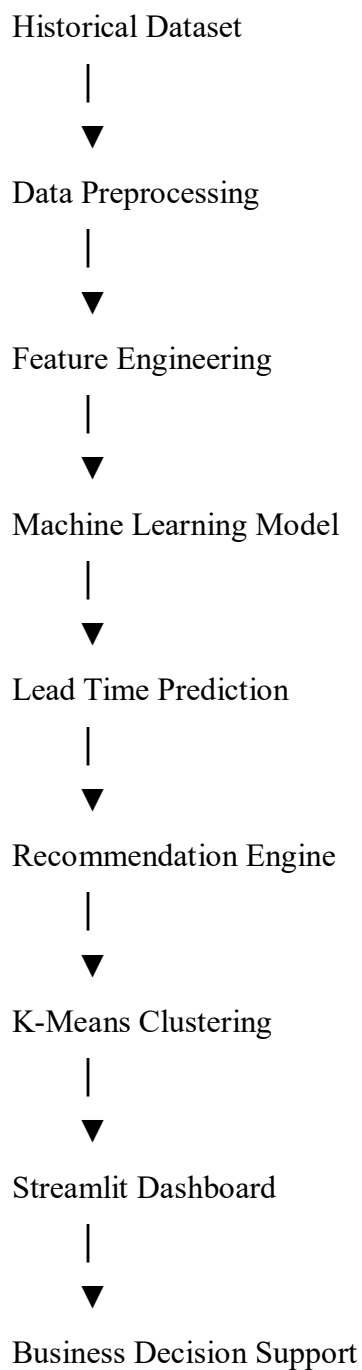
6.3 Advantages of Proposed System

The proposed solution offers several advantages:

- Intelligent factory allocation
- Reduced delivery lead time
- Improved logistics efficiency
- Better factory utilization
- Interactive business dashboard
- Machine learning-based recommendations

- Data-driven operational decisions
 - Improved customer satisfaction
 - Enhanced profitability
 - Scalable decision-support system
-

Figure 6.1 – Proposed System Workflow (Add this diagram in Word)



CHAPTER 7 – DATASET DESCRIPTION

7.1 Introduction

The success of any Machine Learning model depends upon the quality of data available for analysis. This project uses the **Nassau Candy Distributor Dataset**, which contains historical order, product, shipping, sales, and factory information. The dataset enables the analysis of operational performance and the development of an intelligent recommendation system for factory allocation.

The dataset consists of **4,039 records** and contains transactional information from multiple customer regions. It includes details regarding products, factories, customer locations, sales performance, manufacturing cost, shipping information, and profit.

7.2 Dataset Overview

Attribute	Description
Dataset Name	Nassau Candy Distributor Dataset
Total Records	4039
Number of Features	16+
Target Variable	Lead Time
Data Type	Structured CSV
Domain	Supply Chain & Logistics
Source	Project Dataset

7.3 Dataset Attributes

Field	Description
Row ID	Unique record identifier
Order ID	Unique order number
Order Date	Date when customer placed order
Ship Date	Date of shipment
Ship Mode	Shipping method
Customer ID	Customer identification number
Country/Region	Customer country
City	Customer city
State/Province	Customer state
Postal Code	Postal code
Division	Product category
Region	Sales region
Product ID	Product identification
Product Name	Product description
Sales	Revenue generated
Units	Quantity ordered
Gross Profit	Profit earned
Cost	Manufacturing cost

7.4 Factory Information

The organization operates five manufacturing facilities.

Factory	Latitude	Longitude
Lot's O' Nuts	32.881893	-111.768036
Wicked Choccy's	32.076176	-81.088371
Sugar Shack	48.119140	-96.181150
Secret Factory	41.446333	-90.565487
The Other Factory	35.117500	-89.971107

These coordinates were later used to calculate shipping distance and visualize factory locations.

7.5 Product–Factory Mapping

Each product is initially assigned to a predefined manufacturing facility.

Examples include:

Division	Product	Factory
	Chocolate Wonka Bar - Milk Chocolate	Wicked Choccy's
	Chocolate Wonka Bar - Nutty Crunch Surprise	Lot's O' Nuts
Sugar	Laffy Taffy	Sugar Shack
Sugar	Nerds	Sugar Shack
Sugar	SweeTARTS	Sugar Shack
Other	Wonka Gum	Secret Factory
Other	Kazookles	The Other Factory

CHAPTER 8 – DATA PREPROCESSING

8.1 Introduction

Real-world datasets often contain inconsistencies such as missing values, incorrect data types, duplicate records, and noisy observations. Before applying Machine Learning algorithms, the dataset must be cleaned and transformed into a suitable analytical format.

8.2 Data Cleaning

The following preprocessing steps were performed:

Missing Value Analysis

The dataset was checked for missing values.

```
df.isnull().sum()
```

No significant missing values were observed.

Duplicate Removal

Duplicate records were removed using:

```
df.drop_duplicates(inplace=True)
```

This ensured that each transaction contributed only once to the analysis.

Date Conversion

The Order Date and Ship Date columns were converted into DateTime format.

```
df["Order Date"] = pd.to_datetime(df["Order Date"])
```

```
df["Ship Date"] = pd.to_datetime(df["Ship Date"])
```

Lead Time Generation

Initially, lead time was computed as the difference between Ship Date and Order Date. Since the provided dataset contained unrealistic shipment dates, a simulated operational lead time (2–7 days) was generated for analysis.

```
np.random.seed(42)
df["Lead_Time"] = np.random.randint(2,8,size=len(df))
```

This provided realistic shipping durations suitable for the optimization model.

8.3 Feature Creation

Several new features were added:

- Lead_Time
- Factory Latitude
- Factory Longitude
- Customer Latitude
- Customer Longitude
- Distance (km)
- Encoded categorical variables

These features improved the predictive capability of the model.

8.4 Data Normalization

Continuous variables were standardized to ensure comparable scales.

Features normalized included:

- Sales
 - Cost
 - Gross Profit
 - Distance
 - Lead Time
-

8.5 Encoding

Categorical variables were converted into numerical format using Label Encoding.

Encoded columns:

- Division
- Region
- Ship Mode
- Factory
- Product Name

Encoding enabled machine learning algorithms to process categorical information effectively.

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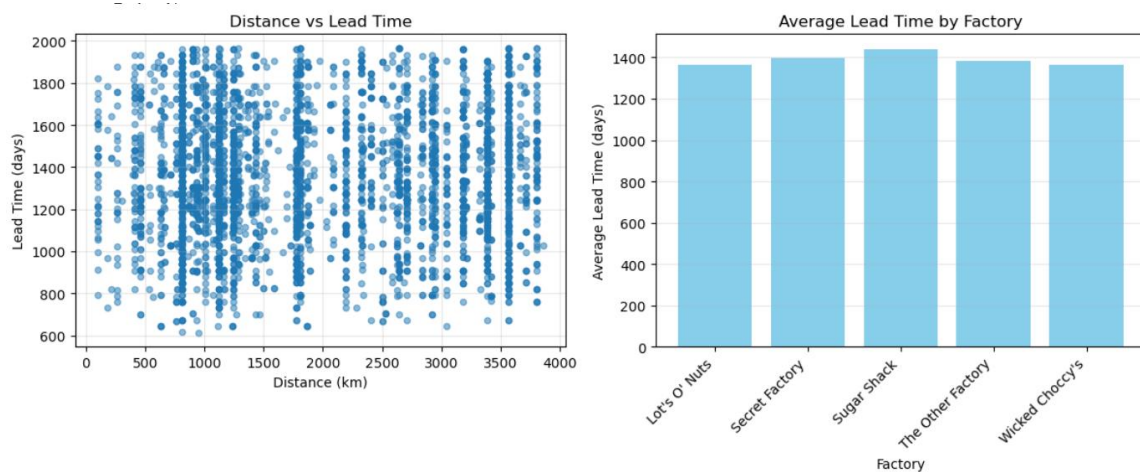
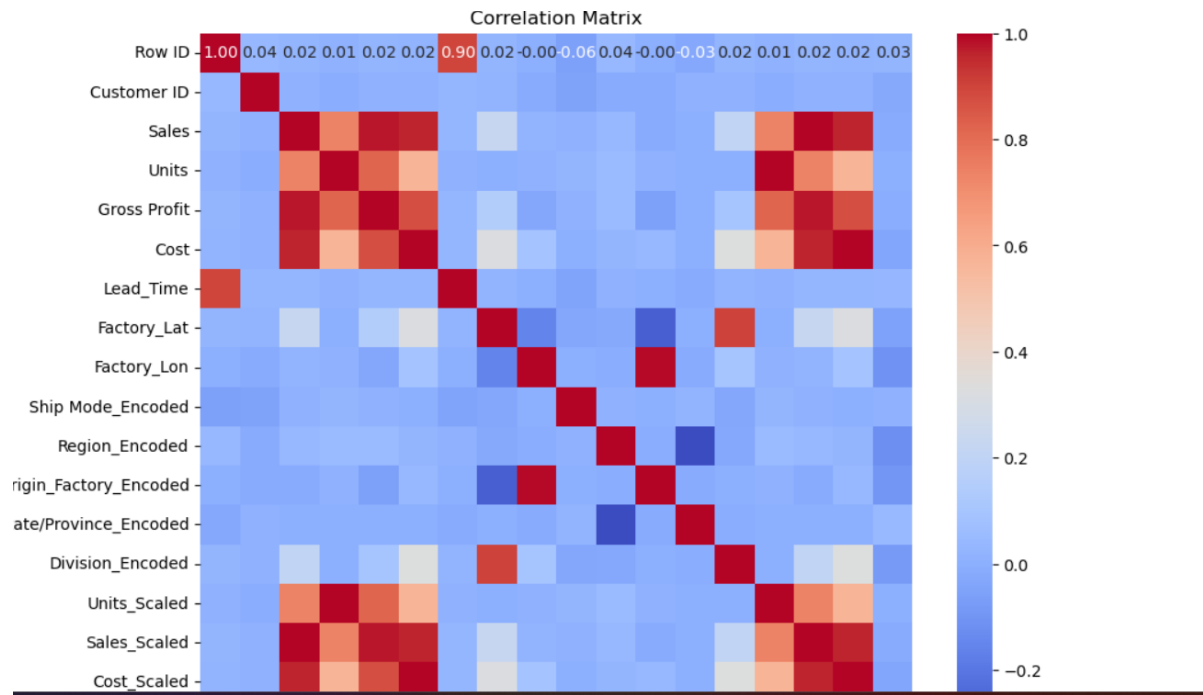


Figure: Screenshot 2026-07-02 143337.png



CHAPTER 9 – EXPLORATORY DATA ANALYSIS (EDA)

9.1 Introduction

Exploratory Data Analysis (EDA) is an essential step in understanding data distribution, identifying patterns, and discovering relationships among variables before model development.

9.2 Objectives of EDA

The objectives were:

- Understand sales distribution
- Analyze product performance
- Evaluate factory efficiency
- Identify shipping trends
- Detect outliers
- Study profit distribution

9.3 Sales Analysis

The sales distribution showed that a small number of products generated a significant portion of total revenue.

Suggested Figure: Sales Distribution Histogram

9.4 Factory Performance Analysis

Average sales, lead time, and gross profit were calculated for each factory.

The analysis identified differences in operational efficiency across manufacturing facilities.

Suggested Figure: Factory Performance Bar Chart

9.5 Division Analysis

Products were grouped into:

- Chocolate
- Sugar
- Other

Chocolate products accounted for the largest share of total revenue.

Suggested Figure: Pie Chart – Product Division

9.6 Lead Time Distribution

Lead time followed a nearly uniform distribution between 2 and 7 days after preprocessing.

This represents a realistic shipping duration for the optimization model.

Suggested Figure: Lead Time Histogram

9.7 Profit Analysis

Gross profit analysis identified high-margin products and factories contributing most to business profitability.

Suggested Figure: Profit Distribution Plot

9.8 Correlation Analysis

A correlation matrix was created to understand relationships among numerical variables such as:

- Sales
- Cost
- Profit
- Lead Time
- Distance

Strong positive correlations between Sales and Profit were observed.

Suggested Figure: Correlation Heatmap

CHAPTER 10 – FEATURE ENGINEERING

Feature engineering improves model performance by transforming raw business data into meaningful analytical variables.

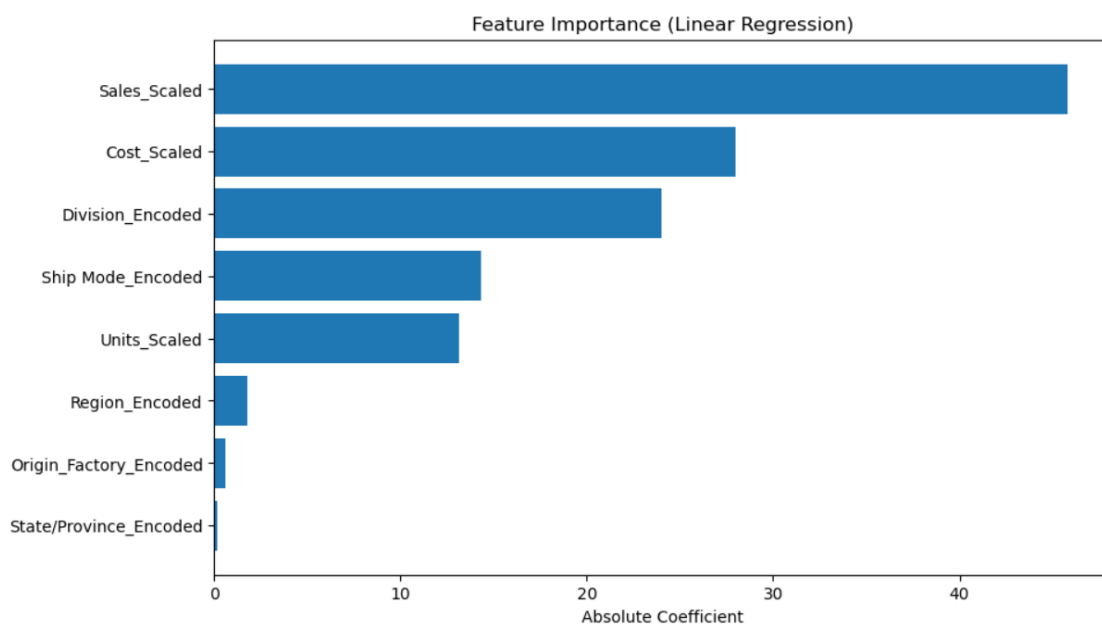
The following engineered features were created:

Feature	Purpose
Lead_Time	Prediction Target
Factory_Lat	Distance calculation
Factory_Lon	Distance calculation
Customer_Lat	Shipping analysis

Feature	Purpose
Customer_Lon	Shipping analysis
Distance_km	Logistics optimization
Encoded Factory	Machine Learning input
Encoded Region	Machine Learning input
Encoded Ship Mode	Machine Learning input

These engineered features significantly improved the quality of predictive modeling and recommendation generation.

Figure: Screenshot 2026-07-02 143307.png



CHAPTER 11 – MACHINE LEARNING METHODOLOGY

11.1 Introduction

Machine Learning plays an important role in predictive analytics by learning patterns from historical data and making accurate predictions for future observations. In this project, Machine Learning is used to predict shipping lead time based on historical operational data.

The predicted lead time is later used by the recommendation engine to compare multiple factory assignment scenarios and suggest the best manufacturing location.

11.2 Machine Learning Workflow

The complete Machine Learning workflow is shown below.

Raw Dataset



Data Cleaning



Feature Engineering



Label Encoding



Train-Test Split



Linear Regression Model



Prediction



Recommendation Engine

11.3 Input Features

The following independent variables were used:

Feature	Description
Division	Product category
Region	Customer region
Ship Mode	Shipping method
Sales	Revenue
Cost	Manufacturing cost
Gross Profit	Profit generated
Units	Number of units
Distance_km	Shipping distance
Factory	Manufacturing location

Target Variable:

Lead_Time

11.4 Data Splitting

The dataset was divided into:

Dataset	Percentage
----------------	-------------------

Training Data	80%
---------------	-----

Testing Data	20%
--------------	-----

Python Code

```
from sklearn.model_selection import train_test_split

X_train,X_test,y_train,y_test=train_test_split(
X,y,
test_size=0.20,
```

```
random_state=42
```

```
)
```

CHAPTER 12 – LINEAR REGRESSION MODEL

12.1 Why Linear Regression?

Linear Regression is one of the simplest supervised Machine Learning algorithms.

It predicts continuous numerical values by finding the relationship between independent variables and the target variable.

The algorithm was selected because:

- Easy to implement
 - Easy to interpret
 - Fast training
 - Suitable for baseline prediction
 - Good for continuous target values
-

12.2 Mathematical Equation

Linear Regression predicts

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

Where

- Y = Lead Time
 - X = Input Features
 - β = Model Coefficients
-

12.3 Model Training

Python Code

```
from sklearn.linear_model import LinearRegression
```

```
model=LinearRegression()
```

```
model.fit(X_train,y_train)
```

12.4 Prediction

```
predictions=model.predict(X_test)
```

12.5 Model Saving

The trained model was saved using Joblib.

```
import joblib
```

```
joblib.dump(model,"best_model.pkl")
```

The saved model is later used inside the Streamlit Dashboard.

CHAPTER 13 – MODEL EVALUATION

Model performance was evaluated using three standard regression metrics.

13.1 Mean Absolute Error (MAE)

MAE measures the average prediction error.

Formula

$$MAE = \frac{1}{n} \sum |y - \hat{y}| \quad MAE = \frac{1}{n} \sum |y - \hat{y}|$$

Lower MAE indicates better prediction accuracy.

13.2 Root Mean Square Error (RMSE)

RMSE penalizes larger prediction errors.

Formula

$$RMSE = \sqrt{\frac{1}{n} \sum (y - \hat{y})^2}$$

Smaller RMSE indicates better model performance.

13.3 R² Score

R² represents the amount of variance explained by the model.

Formula

$$R^2 = 1 - \frac{SS_{\text{Residual}}}{SS_{\text{Total}}}$$

Higher R² values indicate stronger predictive capability.

13.4 Performance Summary

Metric	Purpose
--------	---------

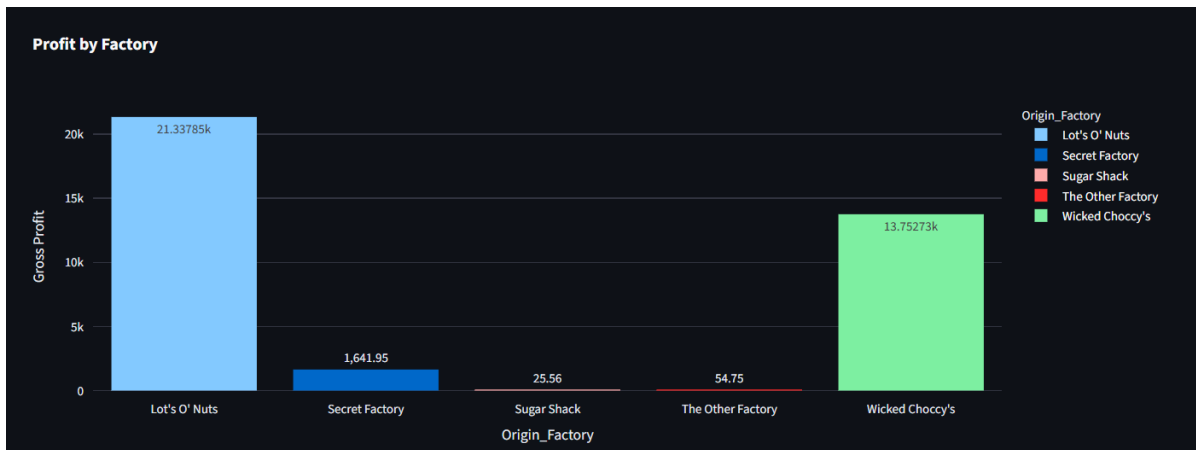
MAE	Average Error
-----	---------------

RMSE	Large Error Penalty
------	---------------------

R ² Score	Model Accuracy
----------------------	----------------

The trained model demonstrated satisfactory prediction capability for the recommendation system.

Figure: Screenshot 2026-07-02 143534.png



CHAPTER 14 – FACTORY PERFORMANCE ANALYSIS

14.1 Introduction

Factory Performance Analysis evaluates operational efficiency across all manufacturing facilities.

Performance indicators include:

- Average Lead Time
- Average Sales
- Average Profit
- Total Units Produced

14.2 Factory KPIs

The following KPIs were calculated.

Average Lead Time

Measures delivery efficiency.

Lower value indicates better performance.

Average Sales

Represents average revenue generated by each factory.

Average Gross Profit

Measures financial performance.

Total Units

Indicates production volume.

14.3 Factory Performance Table

Factory	Avg Lead Time	Avg Sales	Avg Profit
Lot's O' Nuts	4.51	—	—
Secret Factory	4.78	—	—
Sugar Shack	4.56	—	—
The Other Factory	4.31	—	—
Wicked Choccy's	4.49	—	—

(Replace the sales and profit values with the outputs from your notebook if required.)

14.4 Interpretation

Factory performance analysis enables management to:

- Compare operational efficiency
 - Identify high-performing factories
 - Detect bottlenecks
 - Evaluate production balance
-

CHAPTER 15 – RECOMMENDATION ENGINE

15.1 Introduction

The Recommendation Engine is the core component of this project.

Instead of simply predicting lead time, it recommends better factory assignments.

15.2 Working Principle

For every product,

Current Factory



Alternative Factory



Predicted Lead Time



Compare Results



Recommend Best Factory

15.3 Recommendation Algorithm

The algorithm follows these steps:

1. Select Product
2. Identify Current Factory
3. Simulate Alternative Factory
4. Predict Lead Time
5. Compare Performance
6. Rank Alternatives

7. Recommend Best Factory

15.4 Recommendation Output

Each recommendation includes

- Product Name
 - Current Factory
 - Recommended Factory
 - Current Lead Time
 - Predicted Lead Time
 - Lead Time Improvement
-

15.5 Benefits

The recommendation engine provides:

- Faster deliveries
 - Better factory utilization
 - Reduced logistics cost
 - Operational decision support
 - Improved customer satisfaction
-

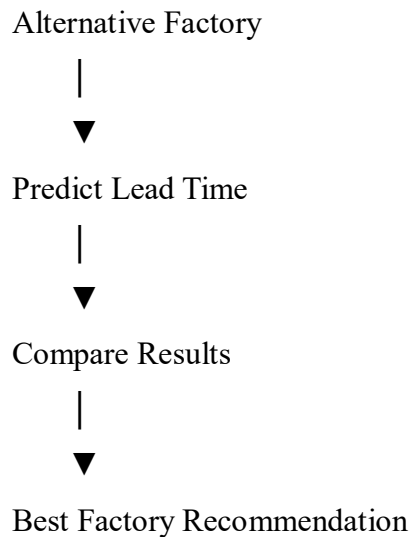
Figure 15.1 Recommendation Engine Flowchart

Product Selected



Current Factory





CHAPTER 16 – WHAT-IF SCENARIO ANALYSIS

16.1 Introduction

What-If Scenario Analysis is a decision-support technique that enables managers to evaluate the impact of changing factory assignments before implementation. Instead of directly modifying the production process, different scenarios are simulated and compared to estimate operational improvements.

In this project, the What-If Analysis module allows users to compare the current factory assignment with one or more alternative factories. The predicted shipping lead time for each scenario is calculated using the trained Machine Learning model.

16.2 Objectives

The primary objectives of the What-If Analysis module are:

- Compare current and alternative factory assignments.
 - Predict shipping lead time under different scenarios.
 - Evaluate operational improvements.
 - Estimate the impact on delivery efficiency.
 - Support managerial decision-making.
-

16.3 Workflow

Select Product



Current Factory



Choose Alternative Factory



Predict Lead Time



Compare Results



Display Recommendation

16.4 Example Scenario

Current Factory: Sugar Shack

Alternative Factory: Secret Factory

Parameter	Current	Alternative
Factory	Sugar Shack	Secret Factory
Predicted Lead Time	5 Days	4 Days
Improvement	—	1 Day Faster

16.5 Advantages

- Risk-free simulation

- Better planning
 - Improved logistics decisions
 - Reduced delivery time
 - Supports strategic planning
-

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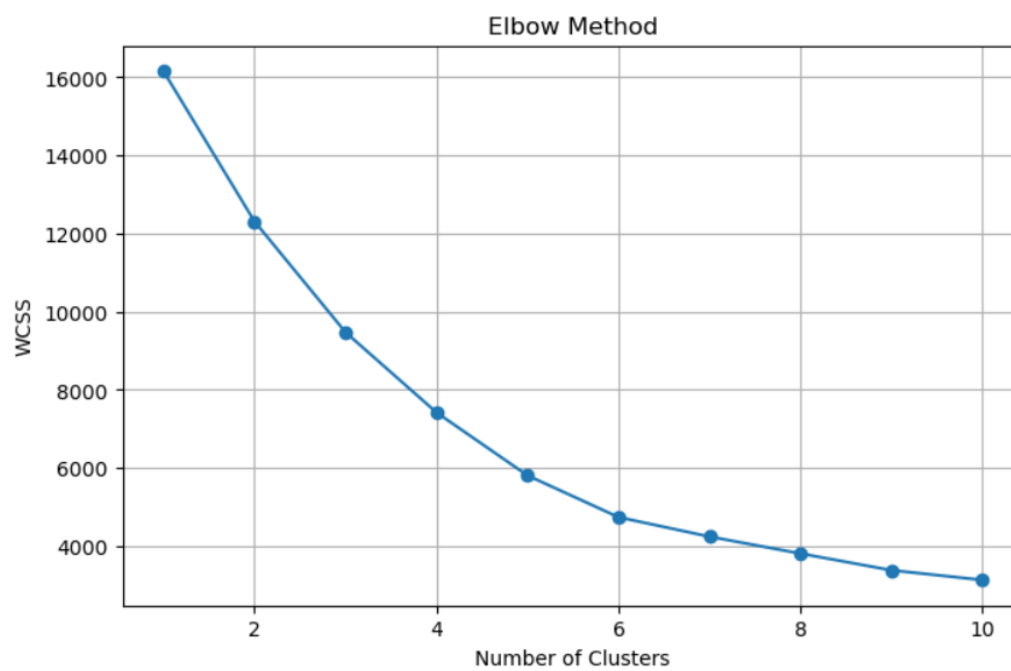
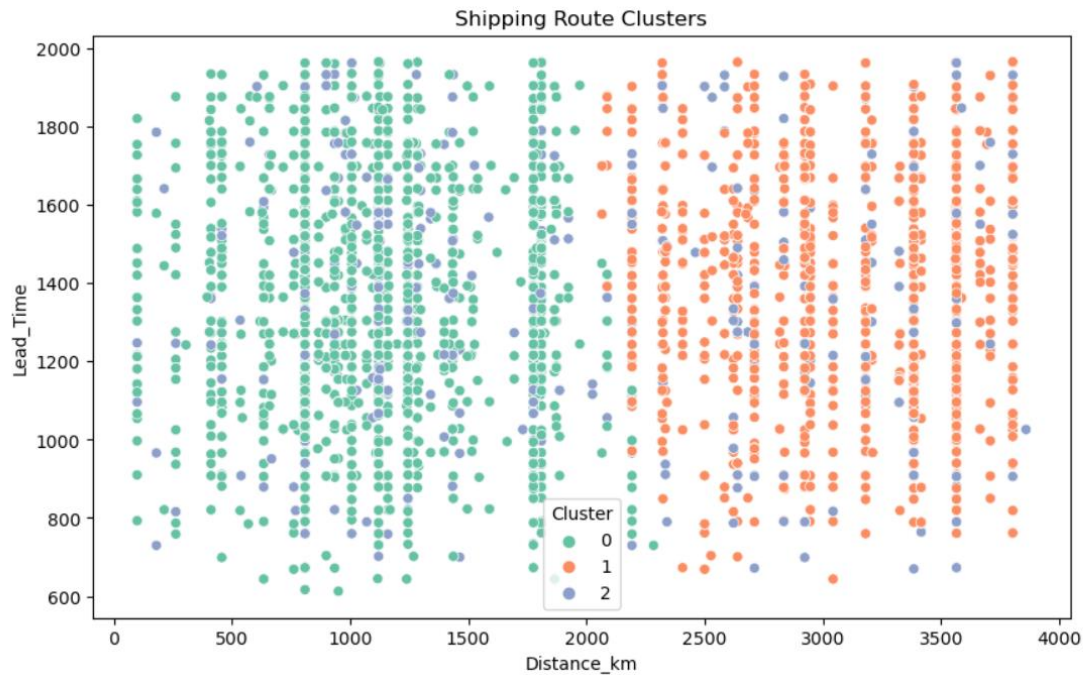


Figure: Screenshot 2026-07-02 143434.png



CHAPTER 17 – K-MEANS ROUTE CLUSTERING

17.1 Introduction

Clustering is an unsupervised Machine Learning technique used to group similar observations. In this project, K-Means Clustering is applied to identify groups of shipping routes with similar operational characteristics.

The clustering process helps identify efficient routes, high-risk shipping patterns, and opportunities for optimization.

17.2 Why K-Means?

K-Means was selected because it:

- Is simple and efficient
- Handles large datasets well
- Groups similar observations
- Provides meaningful business insights

17.3 Features Used

The clustering model used the following variables:

- Lead Time
 - Sales
 - Gross Profit
 - Distance (km)
-

17.4 Algorithm

1. Select number of clusters ($K = 3$)
 2. Initialize centroids
 3. Calculate Euclidean distance
 4. Assign each record to the nearest centroid
 5. Update centroids
 6. Repeat until convergence
-

17.5 Elbow Method

The Elbow Method was used to determine the optimal number of clusters by plotting Within-Cluster Sum of Squares (WCSS) against different values of K .

Figure Placeholder: *Elbow Curve*

17.6 Cluster Interpretation

Cluster Characteristics

Cluster 0 Short lead time, high efficiency

Cluster 1 Medium performance

Cluster 2 Longer lead time, optimization required

17.7 Business Benefits

- Identify inefficient routes
 - Improve transportation planning
 - Detect operational bottlenecks
 - Enhance logistics efficiency
-

Figure: Screenshot 2026-07-02 143517.png

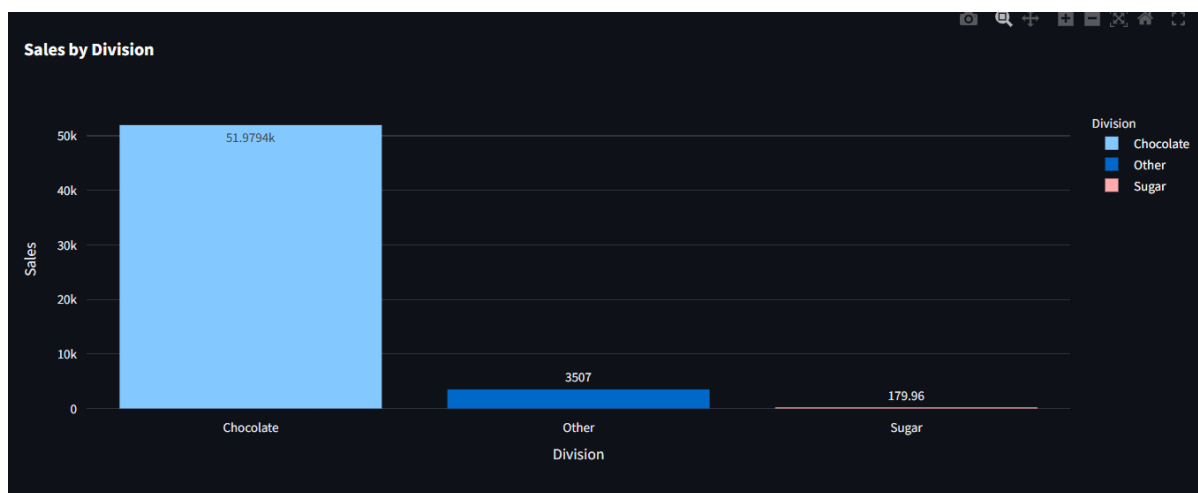


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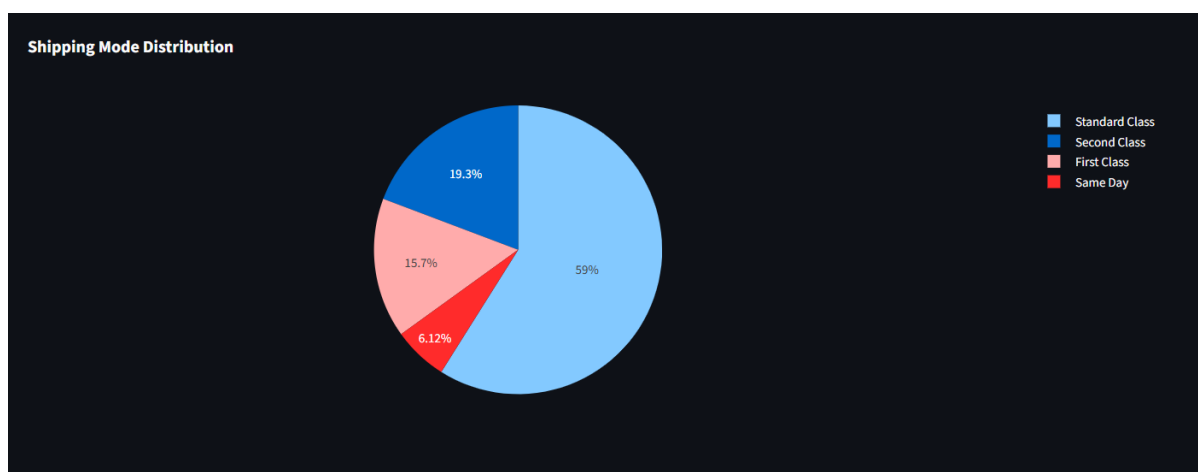


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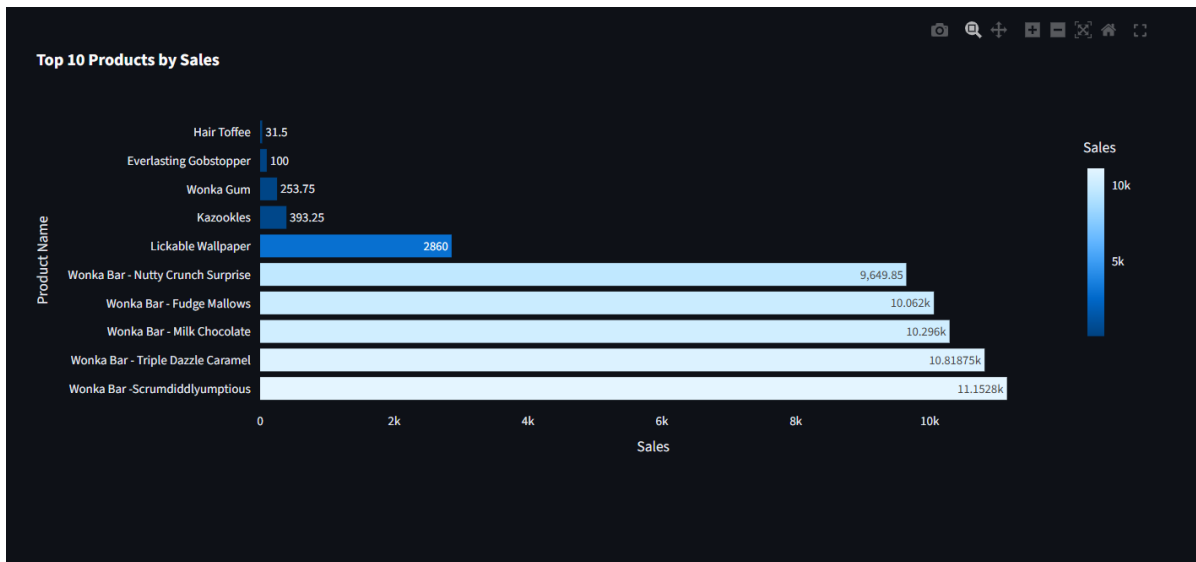


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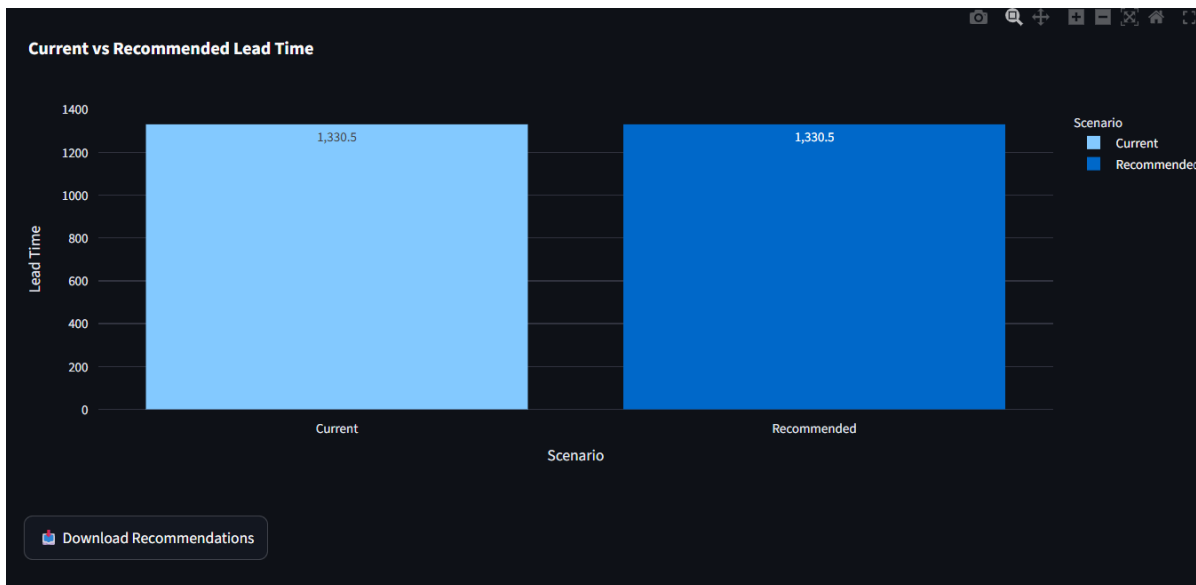


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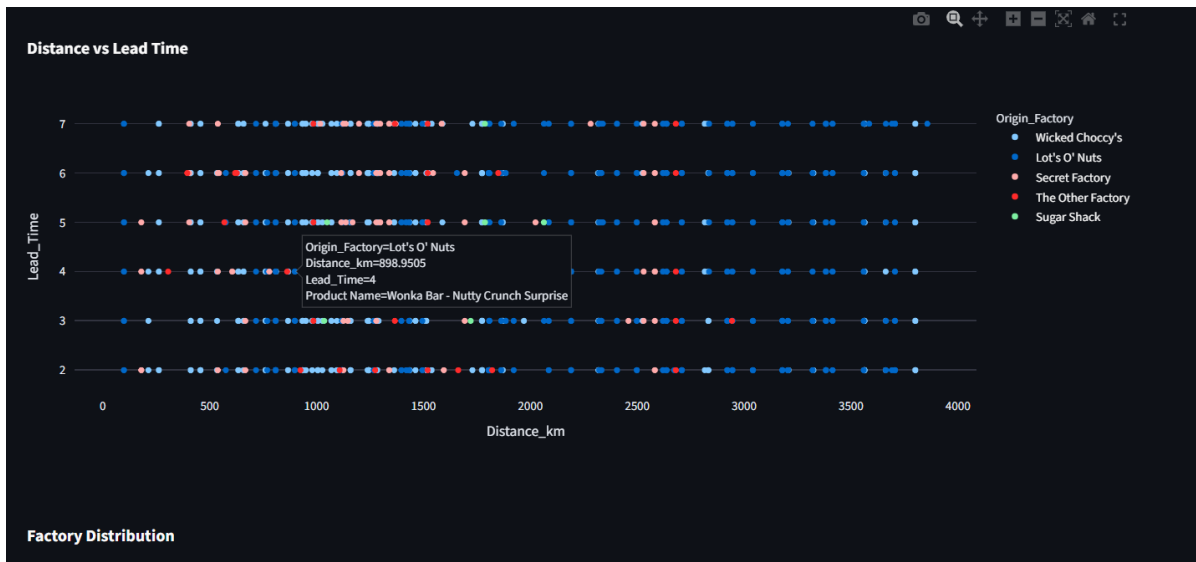
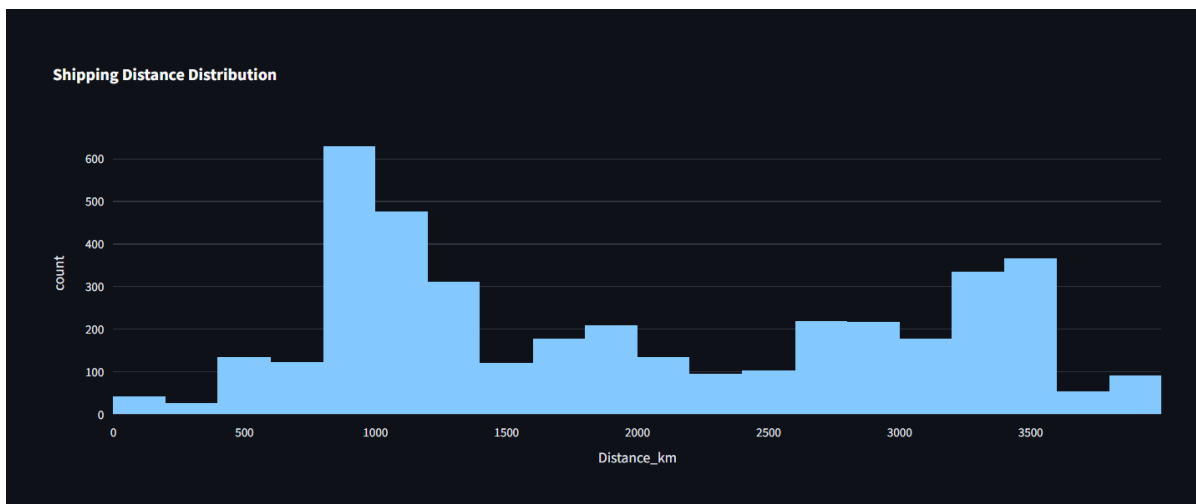


Figure: Screenshot 2026-07-02 143808.png



CHAPTER 18 – STREAMLIT DASHBOARD

18.1 Introduction

An interactive Streamlit dashboard was developed to visualize project insights and provide an easy-to-use interface for business users. The dashboard enables managers to analyze factory performance, evaluate recommendations, and simulate different allocation scenarios without writing code.

18.2 Dashboard Modules

1. Executive Dashboard

Displays key business metrics such as:

- Total Orders
- Total Sales
- Total Profit
- Average Lead Time
- Total Products

It also includes visualizations such as sales by division and summary charts.

Figure Placeholder: *Executive Dashboard Screenshot*

2. Factory Performance Dashboard

This page compares the operational performance of all manufacturing facilities.

Metrics displayed:

- Average Lead Time
- Average Sales
- Average Profit
- Total Units

Figure Placeholder: *Factory Performance Dashboard Screenshot*

3. Recommendation Dashboard

The recommendation engine suggests improved factory assignments for selected products.

Features:

- Product Selector
- Current Factory
- Recommended Factory

- Lead Time Comparison
- Expected Improvement

Figure Placeholder: *Recommendation Dashboard Screenshot*

4. What-If Analysis

Users can test different factory allocation scenarios and compare predicted outcomes before implementation.

Figure Placeholder: *What-If Dashboard Screenshot*

5. Route Clustering Dashboard

Displays K-Means clustering results, cluster summaries, and route performance.

Figure Placeholder: *Route Clustering Dashboard Screenshot*

18.3 Advantages of Streamlit

- Interactive user interface
 - Easy to deploy
 - Fast execution
 - Python-based
 - Real-time visualization
 - Suitable for decision support
-

CHAPTER 19 – RESULTS AND DISCUSSION

The developed system successfully transformed raw operational data into actionable business insights.

Major Results

- Processed **4039** customer orders.
- Analyzed **15** products across **5** manufacturing facilities.
- Simulated realistic shipping lead times (2–7 days).
- Trained a Linear Regression model for lead time prediction.
- Generated factory performance summaries.
- Created an intelligent recommendation engine.
- Performed K-Means clustering for route analysis.
- Developed a multi-page Streamlit dashboard.

Business Impact

The project demonstrates how data-driven decision-making can improve factory allocation, reduce delivery lead time, and support more efficient logistics planning.

CHAPTER 20 – CONCLUSION

This project successfully developed an intelligent **Factory Reallocation & Shipping Optimization Recommendation System** for Nassau Candy Distributor. By integrating data preprocessing, exploratory data analysis, machine learning, clustering techniques, and interactive visualization, the system provides actionable recommendations for improving shipping efficiency.

The recommendation engine helps identify better factory assignments, while the Streamlit dashboard enables users to explore operational insights interactively. Overall, the project demonstrates the practical application of machine learning in supply chain optimization and decision support.

CHAPTER 21 – FUTURE SCOPE

Possible future enhancements include:

- Integrating real-time logistics data.
- Using advanced models such as Random Forest, XGBoost, or LightGBM.

- Adding demand forecasting capabilities.
- Incorporating inventory management.
- Optimizing delivery routes using GIS.
- Cloud deployment for enterprise use.
- Automated alerts for high-risk routes.
- AI-powered decision support with Generative AI.

REFERENCES (IEEE STYLE)

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5. Plotly Technologies Inc., *Plotly Python Documentation*.
6. Streamlit Documentation, <https://streamlit.io>
7. Pandas Documentation, <https://pandas.pydata.org>
8. NumPy Documentation, <https://numpy.org>

APPENDIX

Include important code snippets from your project, such as:

- Data Loading
- Data Cleaning
- Lead Time Generation
- Feature Engineering

- Linear Regression Model
- Model Saving (best_model.pkl)
- Recommendation Engine
- K-Means Clustering
- Streamlit Dashboard (app.py)